

Firas Alshami

Permanent US Resident in Irvine, CA

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Education

California State University, Fullerton

Completed May 2025

Bachelor of Science in Computer Science

Major GPA: 3.92

Work Experience

Lead Research Assistant

June 2024 - July 2025

Supportive Housing Project

- Created C++ programs for ESP32/Arduino devices, reducing device-to-app latency to under 200ms.
- Led a 5-engineer team to design and deploy an Android (Kotlin) + IoT system serving over 30,000 residents, delivering 3 weeks early and contributing to over \$1.2M in secured funding.
- Designed and deployed bluetooth communication and a secure REST API (Express + Node.js) supporting authenticated client access.
- Conducted end-to-end system testing across mobile, IoT, and backend components to ensure reliable real-time device synchronization.

Projects

Go Database | Go, Systems Programming

- Implemented a disk-backed key-value engine in Go using a page-based B-Tree with fixed 4KB pages and custom binary layout.
- Designed insert/lookup with in-node binary search and recursive splitting, and built page allocation for persistent storage.

HTTP Server | C, Socket Programming

- Implemented a HTTP/1.1 file server in C using POSIX sockets, handling request parsing and file streaming over TCP.
- Implemented connection handling using socket, bind, listen, and accept, manually parsing HTTP headers and streaming files in chunks over TCP.

Chess Board Recognition | Python, OpenCV, TensorFlow, React, Node.js

- Developed a full-stack React + Node.js application integrating a TensorFlow model for chessboard image analysis.
- Containerized a FastAPI inference service and achieved ~1.3s end-to-end latency.

Skills

- **Languages:** Go, JavaScript/TypeScript, SQL, Python, C++, C, Kotlin
- **Frameworks/Libraries:** React, Next.js, Node.js, Express, FastAPI, TensorFlow
- **Tools & Platforms:** Git, AWS, Docker